“[A niche] is a highly abstract multi-dimensional hyperspace.”

George Evelyn Hutchinson

The importance of habitat

Ecological niches depend on the existence of a stable habitat; small changes can eradicate niches that organisms once filled. For example, dragonfly larvae only develop within a certain range of water acidity, chemical composition, temperature, and prey, and with a limited number of predators. The right vegetation is needed by adult females for egg-laying, and by larvae for metamorphosis. The dragonfly also impacts its environment: its eggs are food for amphibians; its larvae, which are both predators and prey, add nutrients to the water; and the adults prey on insects. These requirements and impacts define its ecological niche. Hutchinson argued that for a species to persist, conditions had to be within the required ranges. If conditions moved outside the niche requirements, a species could face extinction.